

Mohammad Raihan Uddin

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EDUCATION

Clemson University PhD in Computer Engineering, Jan. 2016 - Present	Clemson, SC, USA
The University of Alabama in Huntsville MSE in Electrical Engineering, Jan. 2024 - Dec. 2025	Huntsville, AL, USA
Daffodil International University BSc in Electronics & Telecommunication Engineering, Jan. 2013 - Feb. 2017	Dhaka, Bangladesh

RESEARCH EXPERIENCE

Graduate Research Assistant - IS-WiN Lab Jan 2026 - Present

- My research focuses on advanced machine learning, artificial intelligence, and wireless communications
- Currently working on ORAN 6G and AI.

Researcher - Networking, Intelligence and Security Lab (NIS Lab) Jan 2024 - Dec 2025

- My research focused on federated learning, artificial intelligence, quantum learning, privacy & security, and wireless communication.
- Key projects include developing a Quantum Federated Learning simulator (SimQFL), designing secure UAV networks with Zero-Knowledge Proofs, and building attack detection systems for smart grids.

TEACHING EXPERIENCE

Graduate Teaching Assistant, ECE Department, UAH Jan 2024 - Dec 2025

- I support and mentor students by delivering course content, facilitating lab sessions, and contributing to curriculum development.
- Courses supported: Computer Organization, Intro to Software Engineering, Intro to Computer Networks, Electric Circuits & Electronics Design Lab, Electric Circuits System, Operating System Lab, Intro to Computer Programming for Engineering Lab.

LEADERSHIP EXPERIENCE

Secretary, Graduate Student Association, UAH Aug 2024 – May 2025

- Managed official correspondence and documentation for the graduate student body, ensuring transparent communication between students and university administration.
- Identified service gaps and collaborated on initiatives to improve graduate student resources and campus engagement.

Head Organizer, TEDxUAHuntsville Jan 2024 – Dec 2025

- Spearheaded the planning and execution of TEDxUAHuntsville 2025, overseeing speaker recruitment, venue logistics, and a multi-disciplinary team of volunteers.
- Curated an intellectual platform for the Huntsville community to share innovative ideas in technology, education, and design.

President, Bangladeshi Student Association, UAH

Aug 2024 – Jan 2026

- Led the executive board in organizing cultural and social events to support international students and foster a sense of community on campus.
- Acted as the primary liaison between the organization and university leadership for funding and event approvals.

President, House of Youth Dialogue (HYD)

Jan 2022 – Dec 2023

- Provided strategic vision for the only international MUN organization originating from Bangladesh, overseeing global youth engagement and leadership programs.
- Directed the organization of international conferences across South Asian countries to equip youth with diplomatic, critical thinking, and public speaking skills.
- Championed social inclusion initiatives to integrate marginalized young people by developing essential life skills through the MUN platform.

General Secretary, House of Youth Dialogue (HYD)

Oct 2019 – Dec 2021

- Managed administrative operations, membership records, and internal communications for a registered charitable organization focused on global education.
- Coordinated logistics for international Model United Nations (MUN) conferences, facilitating intellectual exchange for students across borders.
- Ensured organizational compliance with charitable regulations while maintaining collaborative partnerships with regional youth development agencies.

Active Member & Volunteer, IEEE

2015 – Present

- Contributed to student engagement and conference coordination across the Huntsville Section and Bangladesh Chapter.
- Facilitated networking opportunities and professional development workshops for engineering students across South Asia.

RESEARCH INTERESTS

AI, ML, Wireless Communication

PUBLICATIONS

Published Papers

1. M. B. Zami, **M. R. Uddin**, D. C. Nguyen, "Secure UAV-assisted Federated Learning: A Digital Twin-Driven Approach with Zero-Knowledge Proofs," Submitted to *IEEE Internet of Things Journal*.
2. V. K. Quy, S. Shaon, **M. R. Uddin**, N. M. Quy, D. C. Nguyen, "Aerial Intelligent Reflecting Surface (A-IRS) for Wireless Power Transfer in 6G Networks: Security Challenges and Solutions," Submitted to *IEEE Network Magazine*.

3. S. Shaon, **M. R. Uddin**, D. C. Nguyen, "Quantum Federated Learning in AI-Native 6G Wireless Networks," Submitted to *IEEE Network Magazine*.
4. **M. R. Uddin**, D. C. Nguyen, "Quantum Federated Learning: A Comprehensive Survey," Submitted to *IEEE Communications Surveys & Tutorials*.
5. R. Rahman, A. Pokharel, **M. R. Uddin**, D. C. Nguyen, "SimQFL: A Quantum Federated Learning Simulator with Real-Time Visualization," Submitted to *IEEE Transactions on Quantum Engineering*.
6. R. Rahman, A. Pokharel, **M. R. Uddin**, D. C. Nguyen, "SimQFL: A Quantum Federated Learning Simulator with Real-Time Visualization," in Proceedings of *IEEE International Conference on Quantum Computing and Engineering*, 2025.
7. **M. R. Uddin**, R. Rahman, D. C. Nguyen, "False Data Injection Attack Detection in Edge-based Smart Metering Networks with Federated Learning," in Proceedings of *IEEE Consumer Communications & Networking Conference (CCNC)*, 2025.
8. V. K. Quy, N. M. Quy, T. T. Hoai, S. Shaon, **M. R. Uddin**, D. C. Nguyen, T., & P. Chatzimisios, "From Federated Learning to Quantum Federated Learning for Space-Air-Ground Integrated Networks," in Proceedings of *IEEE Conference on Standards for Communications and Networking (CSCN)*, 2024.
9. M. Arefin, **M. R. Uddin**, N. A. Evan, M. R. Alam, et al., "Enterprise network: Security enhancement and policy management using next-generation firewall (NGFW)," in *Computer Networks, Big Data and IoT*, Springer, 2021, pp. 753–769.
10. **M. Uddin**, N. A. Evan, M. R. Alam, M. Arefin, et al., "Analysis of generic routing encapsulation (GRE) over IPsec VPN tunneling in IPv6 network," in *International Conference on Ubiquitous Communications and Network Computing*, Springer, 2021, pp. 3–15.
11. S. K. Dey, **M. R. Uddin**, M. Rahman, et al., "Performance analysis of SDN-based intrusion detection model with feature selection approach," in *Proceedings of international joint conference on computational intelligence*, Springer, 2020, pp. 483–494.
12. N. F. Firoz, M. Arefin, **M. R. Uddin**, et al., "Performance optimization of layered signature based intrusion detection system using Snort," in *International Conference on Cyber Security and Computer Science*, Springer, 2020, pp. 14–27.
13. **M. R. Uddin**, K. M. Kabir, and M. T. Arefin, "Artificial neural network inducement for enhancement of cloud computing security," in *2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)*, IEEE, 2019, pp. 1–6.
14. S. K. Dey, M. M. Rahman, and **M. R. Uddin**, "Detection of flow based anomaly in OpenFlow controller: Machine learning approach in software defined networking," in *2018 4th International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT)*, IEEE, 2018, pp. 416–421.
15. S. K. Dey, **M. R. Uddin**, K. M. Kabir, and M. M. Rahman, "Enhancing the security of cloud computing: Genetic algorithm and QR code approach," in *2017 4th International Conference on Advances in Electrical Engineering (ICAEE)*, IEEE, 2017, pp. 181–186.
16. **M. R. Uddin**, K. M. Kabir, and M. M. Hasan, "Data security through image processing by Blowfish algorithm, genetic algorithm and LSB," *Foundation of Computer Science*, vol. 148, 2016.

Pre-prints & Under Review

- [17] **M. R. Uddin**, D. C. Nguyen, "Next-Generation UAVs: Survey of Emerging Technologies and Security & Privacy," Submitted to *Vehicular Communications*.
- [18] **M. R. Uddin**, R. Rahman, "Quantum Classical Federated Learning for Intrusion Detection in UAV Communication Networks," Submitted to *IEEE Transactions on Vehicular Technology*.

- [19] R. Rahman, **M. R. Uddin**, “Enabling Intelligent Satellite Communication using Personalized Quantum Federated Learning,” Submitted to *IEEE Transactions on Vehicular Technology* .
- [20] **M. R. Uddin**, F. Afghah, “ORAN-DEFEND: LLM-Augmented Anomaly Reasoning for Open Radio Access Networks,” Submitted to *INFOCOM 2026*.
- [21] **M. R. Uddin**, T. Seyfi, F. Afghah, “RFPrompt: Prompt-Based Expert Adaptation of the Large Wireless Model for Modulation Classification,” Submitted to *IEEE Globecom 2026*.

TECHNICAL SKILLS & CERTIFICATIONS

Programming & Frameworks: Python, MATLAB, C++, PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, Qiskit, PennyLane, TensorFlow Quantum, ZoKrates, zk-STARKs, ZoKrates

Systems & Technical: Cisco (CCNA), RedHat (RHCSA, RHCE), AWS (Solutions Architect), Docker (DCA), Kubernetes (CKA), MySQL, LaTeX, and Linux Systems.

Certifications

Certified Kubernetes Administrator (CKA)
 AWS Certified Solutions Architect – Associate
 Docker Certified Associate (DCA)
 Red Hat Certified Engineer (RHCE)
 Red Hat Certified System Administrator (RHCSA)
 Cisco Certified Network Associate (CCNA)

PROFESSIONAL SERVICE & LEADERSHIP

Head Organizer, TEDxUA Huntsville 2025	2025
Secretary, UAH Graduate Student Association	2024 - 2025
President, UAH Bangladeshi Student Association	2024 - 2026
President, House of Youth Dialogue	2022 - 2023
General Secretary, House of Youth Dialogue	2019 - 2021
Graduate Student Member, IEEE (Communication and Computer Societies)	2015 -
Volunteer, IEEE-USA IWRC Aerospace & Defense	2024

Peer Review Service

Served as a peer reviewer for nearly 30 papers for the following top-tier journals:

- *IEEE Internet of Things Journal*
- *ACM Computing Surveys*
- *IEEE Open Journal of the Communications Society*
- *IEEE Internet of Things Magazine*
- *IEEE Transactions on Green Communications and Networking*
- *IEEE Transactions on Network Science and Engineering*
- *IEEE Transactions on Cognitive Communications and Networking*
- *IEEE Networks*
- *INFOCOM NetRobiCS 2026*
- *Computer Networks*
- *IEEE Transactions on Dependable and Secure Computing*

- *IEEE Communications Letters*

REFERENCES

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